

Geometry-Aware Line Graph Transformer Pretraining for Molecular Property Prediction

Peizhen Bai¹, Xianyuan Liu², Wenrui Fan², Tianyi Jiang², William K. Cheung³, *Member, IEEE*,
and Haiping Lu², *Senior Member, IEEE*

Abstract—Molecular property prediction with deep learning approaches has gained much attention over the past years. Due to the scarcity of labeled molecules, there has been growing interest in self-supervised learning (SSL) methods that learn generalizable molecular representations from unlabeled data through pretraining. Molecules are typically modeled as 2-D topological graphs. However, their 3-D geometry is also important in determining molecular functionalities, and such 3-D information can enhance 2-D molecular representation learning. The 2-D–3-D pretraining has two key challenges: 1) how to construct a high-capacity backbone network to encode the dual-modality molecular information and 2) how to design effective pretraining tasks to learn both intermodality and intramodality relations simultaneously. To tackle these two challenges, we propose a Geometry-aware line graph transformer (Galformer) pretraining framework, a novel SSL method that aims to enhance molecular representation learning from a dual-modality perspective. Specifically, we first design a dual-modality line graph transformer backbone to adaptively encode a molecule’s 2-D topological and 3-D geometric line graphs. The designed backbone has a high capacity and can capture critical structural information from both 2-D and 3-D modalities. Then, we devise two complementary pretraining tasks to achieve a comprehensive understanding at both intermodality and intramodality levels. These tasks provide properly supervised information and effectively extract discriminative 2-D and 3-D knowledge from unlabeled molecules. We evaluate Galformer against ten state-of-the-art baselines on 15 property prediction benchmarks via downstream fine-tuning. The experimental results show that Galformer consistently outperforms all baselines on both classification and regression tasks, demonstrating its effectiveness. Our source code is available at GitHub: <https://github.com/peizhenbai/Galformer>

Index Terms—Graph transformer, molecular property prediction, multimodal learning, self-supervised learning (SSL).

I. INTRODUCTION

PREDICTING molecular properties plays a crucial role in computational chemistry and drug discovery. Traditional feature-based methods, such as molecular fingerprints [1] and descriptors [2], are widely used in these fields but suffer from data sparsity and overreliance on prior knowledge. Recent advances in deep learning have shifted the focus to data-driven representation learning [3]. Using large-scale molecular data, deep representation learning can potentially generate more comprehensive molecular representations that can be advantageous for various property prediction tasks [4], [5], [6], [7], [8].

Due to the enormous chemical space, only a tiny fraction of molecules are labeled, making supervised deep learning methods prone to overfitting and poor generalization to out-of-distribution molecules [9], [10]. Motivated by the prominent success of self-supervised learning (SSL) in computer vision and natural language processing, recent studies have applied the standard SSL paradigm to molecular representation learning and prediction. In this paradigm, a model is first pretrained on large-scale unlabeled molecules, and then fine-tuned to different downstream tasks with limited molecular labels. SSL has proven effective in capturing molecular information and improving predictive performance [11], [12], [13], [14].

In recent years, most molecular representation learning methods within the SSL framework have utilized graph neural networks (GNNs) for encoding 2-D molecular graphs, where atoms are represented as nodes and chemical bonds as edges. Through the message-passing mechanism, GNNs can effectively preserve substantial topological information in molecules [15]. One line of these works focuses on developing graph contrastive learning for molecular representation learning, which adopts data augmentation strategies to generate augmented 2-D graphs from the same molecule [16], [17], [18], [19]. However, most augmentation strategies, such as node dropping and bond deletion, could potentially disturb the important chemical structure of a molecule [18], [20], [21]. On the other hand, these previous works have only considered the 2-D modality of molecular data, despite its multimodal nature. More recently, there has been a growing trend toward

Received 13 October 2024; revised 20 May 2025 and 4 December 2025; accepted 19 May 2026. This work was supported in part by the Engineering and Physical Sciences Research Council under Grant UKRI396 and in part by the Institute of Systems Medicine and Health Sciences, Hong Kong Baptist University. (*Corresponding author: Haiping Lu.*)

Peizhen Bai is with the School of Computer Science, University of Sheffield, S10 2TN Sheffield, U.K. (e-mail: peizhen.bai@sheffield.ac.uk).

Xianyuan Liu, Wenrui Fan, and Haiping Lu are with the School of Computer Science and the Centre for Machine Intelligence, University of Sheffield, S10 2TN Sheffield, U.K. (e-mail: xianyuan.liu@sheffield.ac.uk; wenrui.fan@sheffield.ac.uk; h.lu@sheffield.ac.uk).

Tianyi Jiang is with the Institute of Cyberspace Security, College of Information Engineering, Zhejiang University of Technology, Hangzhou 310023, China, and also with the School of Computer Science and the Centre for Machine Intelligence, University of Sheffield, S10 2TN Sheffield, U.K. (e-mail: josieyi0319@163.com).

William K. Cheung is with the Department of Computer Science, Hong Kong Baptist University, Hong Kong, China (e-mail: william@comp.hkbu.edu.hk).

Digital Object Identifier 10.1109/TNNLS.2026.3698579

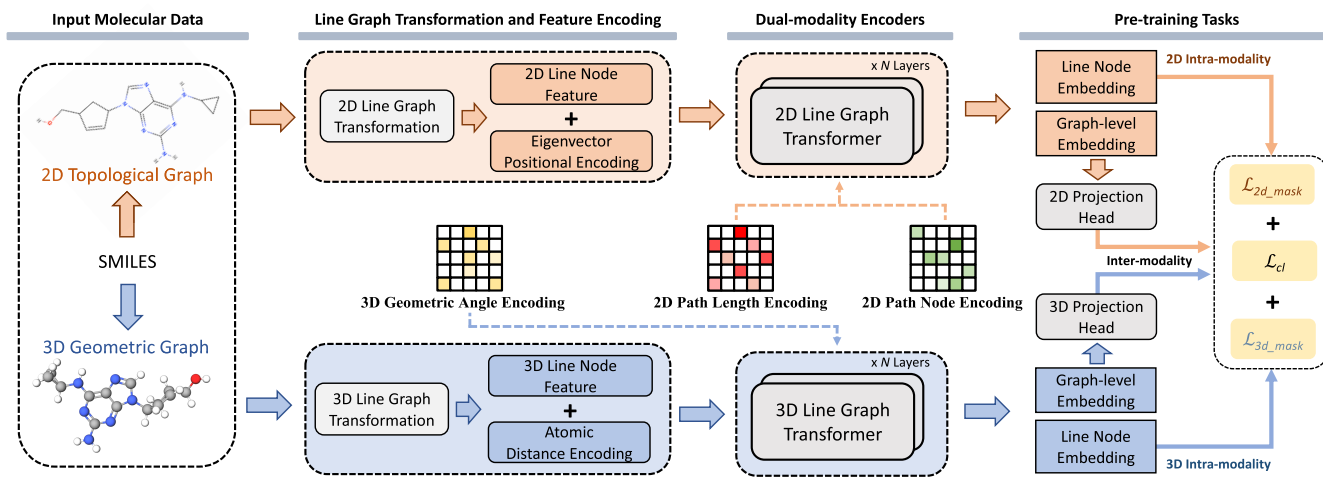


Fig. 1. Overview of the Galformer pretraining framework. The input 2-D topological graph and 3-D geometric graph are derived from the same molecule in the SMILES [30] format. Galformer provides a dual-modality line graph transformer architecture to encode both 2-D and 3-D molecular information. In each modality, the original molecular graph is first transformed into its line graph. Then, the line node features and positional/distance encoding are derived. Furthermore, the 2-D and 3-D encoders learn structural information by incorporating 2-D path length, 2-D path node, and 3-D geometric angle encodings in their self-attention modules. For pretraining tasks, Galformer creates the intermodality mask on line node embedding, and the intramodality contrast on graph-level embedding.

integrating 3-D geometric information into 2-D molecular representation learning and designing new contrastive methods across modalities [22], [23], [24], [25]. Incorporating 3-D geometry can provide rich additional knowledge vital in determining molecular functionalities [26].

Despite the fruitful developments, two challenges remain in 2-D–3-D molecular SSL: limited model capacity and incomplete pretraining tasks. On the one hand, previous studies primarily employ existing 2-D and 3-D GNNs as backbone networks for pretraining, which have consistently suffered from the over-smoothing problem and expressive power limitation [27]. These issues limit the model capacity to capture rich multimodal information and learn expressive representations from molecular data. On the other hand, most dual-modality molecular pretraining methods only focus on learning intermodality relations between 2-D and 3-D molecular structures without simultaneous consideration of intramodality relations. The intramodality relation within each modality complements the intermodality relation [28], [29]. For instance, each 2-D molecular representation should not only obtain geometric information from its associated 3-D modality, but also preserve topological information within the 2-D structure. By considering both intermodality and intramodality knowledge extraction in a unified molecular SSL framework, pretrained models can be more robust and generalizable to various downstream tasks.

To tackle the two challenges, we propose the geometry-aware line graph transformer (Galformer), a unified 2-D–3-D pretraining framework for learning generalizable molecular representations. Fig. 1 presents an overview of this framework. We first transform the input 2-D and 3-D molecular graphs into line graphs (i.e., edge-adjacency graphs) individually. This transformation preserves the inherent adjacency information in molecules while emphasizing structural information at finer granularity levels, such as node-pair distances and edge-pair angles. Subsequently, we design a dual-modality line graph transformer architecture to model both 2-D and

3-D molecular information flow simultaneously. To effectively consider the intermodality and intramodality relations in SSL, we design two complementary pretraining tasks: masked line node prediction and dual-modality contrastive learning. At the intramodality level, we randomly mask a proportion of 2-D and 3-D line nodes, and then predict their original types from the context embedding within the individual modality. Due to the property of the line graph, this strategy considers local atom-pair, bond, and angle information being masked, thus capturing richer structural patterns in molecules. At the intermodality level, we apply contrastive learning to maximize the mutual information between the graph-level 2-D and 3-D molecular representations. This contrastive pretraining task bridges the gap between modalities and complements intermodality mask learning. Finally, we propose the pretrained 2-D encoder, enriched with 3-D geometric knowledge, to enhance downstream property prediction. Our contributions can be summarized as follows.

- 1) We propose a dual-modality line graph transformer architecture to encode both 2-D topological and 3-D geometric information of molecules. By transforming molecular graphs into line graphs and introducing structural encodings as inductive bias into the transformer, the proposed architecture achieves an improvement in model capacity and in capturing discriminative knowledge across modalities.
- 2) We design two complementary pretraining tasks: masked line node prediction at the intermodality level and dual-modality contrastive learning at the intramodality level. By jointly modeling both intramodality and intermodality relations in a unified framework, our approach provides a complete and robust pretraining objective for molecular representation learning.
- 3) We evaluate Galformer performance on 15 downstream molecular property datasets, including both classification

and regression tasks. The experimental results show its superiority over ten state-of-the-art baselines.

The remainder of this article is organized as follows. Section II provides a brief overview of related works. Section III details the preliminaries and relevant definitions. Our proposed framework is introduced in Section IV. Section V presents the experimental details and comparative analysis. Finally, conclusions are given in Section VI.

II. RELATED WORKS

A. Molecular Representation Learning

As the critical foundation of property prediction, many studies have been devoted to improving molecular representation learning. Traditional feature-based methods depend on hand-crafted representations generated by chemical fingerprints or molecular descriptors, which lack the ability to capture potential structural information and require extensive feature engineering [1], [2], [31], [32]. In recent years, the emergence of GNNs has allowed for more intuitive and efficient learning of molecular graph representations. For instance, Gilmer et al. [33] first proposed to use a message-passing framework to capture atomic interactions in molecular graphs, and then Yang et al. [6] extended the framework by considering the directional bond interactions. Xiong et al. [7] leveraged graph attention networks to learn the data-driven fingerprints in a dense vector space. Nevertheless, these approaches still face challenges in handling the geometry of molecules and capturing long-range dependencies, which has led to the exploration of transformer-based architectures to preserve structural information and improve the generalization of learned molecular representations [34], [35], [36], [37], [38]. Furthermore, advances in general graph learning also provide valuable insights for molecular representation. For instance, HAQJSK [39] demonstrates strong performance in graph classification tasks, while PEGFAN [40] addresses the challenges of modeling heterophilous graphs. These developments offer complementary perspectives that can inform the design of more expressive molecular graph models. ALIGNN [41] performs iterative local message-passing on both the original atomistic bond graph and its 3-D line graph for materials property prediction. In contrast, our approach emphasizes a global receptive field through geometry-aware attention biases and transferable molecular representations derived from dual-modality line graphs. In addition, Li et al. [42] presented a comprehensive survey of recent theoretical and algorithmic progress, underscoring the importance of structure-aware modeling in this domain.

B. Molecular SSL

SSL has achieved fruitful progress in the area of molecular representation learning, leveraging the potential of unlabeled molecular data. Many pretraining tasks have been developed and can be broadly classified into contrastive methods, generative/predictive methods, or a combination of them [43], [44], [45], [46]. The principal objective of contrastive methods is to maximize the mutual information between the augmented perspectives from the same molecular graph. These

approaches suggest different augmentation strategies, such as atom masking and bond deletion, to create semantically similar perspectives [18], [19], [20], [47], [48], [49], [50]. In contrast, generative/predictive methods consider reconstructing the information of a single-molecular graph at different levels, or creating certain pseudo-labels for pretraining tasks [10], [11], [12], [13], [51], [52]. For example, Zhang et al. [11] proposed a framework to capture molecular information from both node-level and graph-level fragmentations, which predicts masked atom-/bond-based attributes and reconstructs motif-based graph trees. Yang et al. [46] employed both contrastive learning with masked atoms and pseudo-label prediction in pretraining. The aforementioned methods are mainly developed to learn representations from 2-D topology molecular graphs exclusively, ignoring the significance of 3-D geometry to property prediction. More recently, a few studies have focused on integrating geometric knowledge into the pretraining framework to obtain more informative representations [21], [22], [51], [52], [53], [54]. In particular, Stärk et al. [22] and Liu et al. [25] propose to encode the 2-D topology and 3-D geometry by different GNN backbone models, and then implicitly inject 3-D information into 2-D GNN with the contrastive and generative methods. Different from these works, we first design the structurally consistent 2-D and 3-D line graph transformers to encode the dual-modality molecular data. Then, we consider the pretraining objective from the multimodal perspective, which learns representations from the inter and intramodality relations simultaneously. Our framework can produce more expressive and generalizable representations for molecular property prediction.

III. PRELIMINARIES

In this section, we briefly recap the concept of transformers [55], the definition of 2-D and 3-D molecular graphs, and problem formulation.

A. Transformer

A vanilla transformer architecture comprises stacked transformer layers, each of which consists of a multihead self-attention module followed by a feedforward network. Both modules embed layer normalization [56] and skip connection [57] to reduce the internal covariate shift and to avoid the vanishing gradient problem, respectively. Let $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n]^T \in \mathbb{R}^{n \times d}$ be an input feature matrix, where d is the hidden dimension, and $\mathbf{x}_i \in \mathbb{R}^d$ is the representation vector of the i th input token. The self-attention module first projects $\mathbf{X}$ into the three representations $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ with the corresponding learnable projection matrices $\mathbf{W}_q \in \mathbb{R}^{d_k \times d}$, $\mathbf{W}_k \in \mathbb{R}^{d_k \times d}$, and $\mathbf{W}_v \in \mathbb{R}^{d_v \times d}$. Then, the attention output is calculated using the scaled dot-product self-attention method, which can be written as

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_q^T, \quad \mathbf{K} = \mathbf{X}\mathbf{W}_k^T, \quad \mathbf{V} = \mathbf{X}\mathbf{W}_v^T \quad (1)$$

$$\mathbf{A} = \frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}} \quad (2)$$

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}(\mathbf{A})\mathbf{V} \quad (3)$$

where $\mathbf{A} \in \mathbb{R}^{n \times n}$ is an attention weight matrix and each element A_{ij} indicates the importance of the j th position while computing the output representation of the i th position in (3). Assume there are m attention heads in the multihead self-attention module; then, the output matrix can be denoted as

$$\begin{aligned} \text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) \\ = \text{Concat}(\text{head}_1, \dots, \text{head}_m) \mathbf{W}_o \\ \text{head}_i = \text{Attention}\left(\mathbf{X}(\mathbf{W}_q^i)^T, \mathbf{X}(\mathbf{W}_k^i)^T, \mathbf{X}(\mathbf{W}_v^i)^T\right) \end{aligned} \quad (4)$$

where $\mathbf{W}_q^i \in \mathbb{R}^{d_k \times d}$, $\mathbf{W}_k^i \in \mathbb{R}^{d_k \times d}$, and $\mathbf{W}_v^i \in \mathbb{R}^{d_v \times d}$ are the projection matrices of the i th head, $\mathbf{W}_o \in \mathbb{R}^{md_v \times d}$ is the multihead output projection matrix, and $\text{Concat}(\cdot)$ denotes the concatenation operator.

Definition 1 (2-D Topological Graph): Given a 2-D molecule, its topological graph can be defined as $\mathcal{G}^{2d} = (\mathcal{V}^{2d}, \mathcal{E}^{2d})$, where each node $u \in \mathcal{V}^{2d}$ denotes an atom and each edge $(u, v) \in \mathcal{E}$ represents a chemical bond between node (atom) u and v . Meanwhile, we initialize the node feature with atom attributes as $\mathbf{h}_u^{2d} \in \mathbb{R}^{\Theta_v}$ and edge feature with chemical bond attributes as $\mathbf{e}_{uv}^{2d} \in \mathbb{R}^{\Theta_e}$, where Θ_v and Θ_e denote the dimensions of initial node and edge features, respectively.

Definition 2 (3-D Geometric Graph): A geometric graph of 3-D molecule can be defined as $\mathcal{G}^{3d} = (\mathcal{V}^{3d}, \mathcal{E}^{3d}, \mathcal{A}^{3d})$, where $(u, v, w) \in \mathcal{A}$ denotes the geometric angle between edge (u, v) and (v, w) . Similar to previous studies [21], [58], we consider the atomic distance $d_{uv} > 0$ and angle $\theta_{uvw} \in [0, \pi]$ as invariant spatial features in $\mathcal{G}^{3d}$, regardless of how the same molecular conformation rotates or translates in 3-D space. To emphasize the geometric information, each 3-D node feature is denoted as a one-hot vector $\mathbf{h}_u^{3d} \in \mathbb{R}^{\Theta_t}$, where Θ_t is the number of atomic types without additional topological attributes.

1) Problem Formulation: Given a set of unlabeled molecules $\mathcal{M} = \{\mathcal{G}_i^{2d}, \mathcal{G}_i^{3d}\}_{i=1}^{|\mathcal{M}|}$, where each molecule $M_i \in \mathcal{M}$ has its 2-D topological graph $\mathcal{G}_i^{2d}$ and 3-D geometric graph $\mathcal{G}_i^{3d}$. The study aims to pretrain a dual-modality SSL model that generates discriminative molecular representations and then adapts to molecular property prediction by fine-tuning. Due to the scarcity of 3-D conformers in downstream tasks, following the paradigm of previous works [22], [25], we regard the knowledge of 3-D geometry as privileged information only used during pretraining. Then, the pretrained 2-D encoder is fine-tuned for downstream property prediction tasks, relying only on the available 2-D topological graphs.

IV. METHODOLOGY

This section presents the details of our proposed pretraining framework, Galformer. First, we describe the procedure for transforming 2-D and 3-D molecular graphs into their corresponding line graphs. Then, we elaborate on the design of dual-modality line graph transformers that can effectively encode both topological and geometric information. Finally, we introduce the pretraining strategy that enables learning of intermodality and intramodality relations.

A. Molecular Line Graphs

1) Topological Line Graph: Given a 2-D topological graph $\mathcal{G}^{2d} = (\mathcal{V}^{2d}, \mathcal{E}^{2d})$, we can transform it into a topological line

graph $\hat{\mathcal{G}}^{2d} = (\hat{\mathcal{V}}^{2d}, \hat{\mathcal{E}}^{2d})$ following the following two steps. First, for each edge in $\mathcal{G}^{2d}$, create a node in $\hat{\mathcal{G}}^{2d}$. This step involves mapping the edges in the original graph to nodes in the line graph. Second, for every pair of edges in $\mathcal{G}^{2d}$ that share a common node, create an edge between their corresponding nodes in $\hat{\mathcal{G}}^{2d}$. This step involves making connections between the nodes in the line graph based on the common nodes in the original graph. After the transformation, we can initialize each node feature in $\hat{\mathcal{G}}^{2d}$ with the atom and bond attributes as follows:

$$\mathbf{n}_{uv}^{2d} = \text{Concat}(\mathbf{W}_g \mathbf{h}_u^{2d} + \mathbf{W}_g \mathbf{h}_v^{2d}, \mathbf{W}_e \mathbf{e}_{uv}^{2d}) \quad (5)$$

where $\mathbf{W}_g \in \mathbb{R}^{\frac{\Theta_v}{2} \times \Theta_v}$ and $\mathbf{W}_e \in \mathbb{R}^{\frac{\Theta_e}{2} \times \Theta_e}$ are learnable projection matrices. To avoid ambiguity and simplify notations, we denote $\hat{\mathcal{V}}^{2d} = \{\hat{v}_i\}_{i=1}^{|\hat{\mathcal{V}}^{2d}|}$ as the set of nodes in topological line graph $\hat{\mathcal{G}}^{2d}$, where $|\hat{\mathcal{V}}^{2d}|$ is equivalent to the number of edges $|\mathcal{E}^{2d}|$ in $\mathcal{G}^{2d}$ by the definition of line graph.

2) Geometric Line Graph: A geometric line graph $\hat{\mathcal{G}}^{3d} = (\hat{\mathcal{V}}^{3d}, \hat{\mathcal{E}}^{3d})$ can be derived from its corresponding 3-D geometric graph $\mathcal{G}^{3d}$ by the similar transformation steps above. In contrast, we emphasize integrating the invariant spatial features (i.e., atomic distance and angle) within the line graph framework. As the atomic distance d_{uv} and angle θ_{uvw} are continuous scalar variables, it is difficult to directly incorporate them into the subsequent deep architecture. Therefore, we use a Gaussian basis kernel function [59] to map the atomic distance and angle into high-dimensional vectors, which can be written as

$$\begin{aligned} \mathbf{d}_{uv} &= \frac{M}{m=1} \left(-\frac{1}{\sqrt{2\pi}|\sigma_d^m|} \exp \left(-\frac{1}{2} \left(\frac{\alpha_d^t d_{uv} + \beta_d^t - \mu_d^m}{|\sigma_d^m|} \right)^2 \right) \right) \\ \theta_{uvw} &= \frac{M}{m=1} \left(-\frac{1}{\sqrt{2\pi}|\sigma_\theta^m|} \exp \left(-\frac{1}{2} \left(\frac{\alpha_\theta^t \theta_{uvw} + \beta_\theta^t - \mu_\theta^m}{|\sigma_\theta^m|} \right)^2 \right) \right) \end{aligned} \quad (6) \quad (7)$$

where $\sim$ denotes the concatenation operator over multiple scalars, and M is the number of Gaussian basis kernels. (σ_d^m, μ_d^m) and $(\sigma_\theta^m, \mu_\theta^m)$ are the m th learnable kernel center and scaling factor for the atomic distance and angle, respectively. (α_d^t, β_d^t) and $(\alpha_\theta^t, \beta_\theta^t)$ are the learnable scalars but indexed by the type of atomic pair in (u, v) and edge pair in (u, v, w) , respectively. The mapped pairwise encodings can preserve invariant spatial features and adapt to further deep modeling. Next, the node and edge features in $\hat{\mathcal{G}}^{3d}$ are defined

$$\mathbf{n}_{uv}^{3d} = \text{Concat}(\mathbf{W}_l \mathbf{h}_u^{3d} + \mathbf{W}_l \mathbf{h}_v^{3d}, \mathbf{W}_d \mathbf{d}_{uv}) \quad (8)$$

$$\mathbf{e}_{uvw}^{3d} = \mathbf{W}_p \theta_{uvw} \quad (9)$$

where $\mathbf{W}_l \in \mathbb{R}^{\frac{\Theta_v}{2} \times \Theta_t}$, $\mathbf{W}_d \in \mathbb{R}^{\frac{\Theta_e}{2} \times M}$, and $\mathbf{W}_p \in \mathbb{R}^{\Theta_e \times M}$ are the learnable projection matrices. By definition, the edges in the line graph represent the pairs of edges in the original graph sharing a common node (i.e., adjacent edges). The relation derives that each line edge corresponds to a unique angle in the original graph. Therefore, we can adopt the projected angle vector $\mathbf{e}_{uvw}^{3d}$ as edge feature in $\hat{\mathcal{G}}^{3d}$. Finally, we define $\hat{\mathcal{V}}^{3d} = \{\hat{v}_i\}_{i=1}^{|\hat{\mathcal{V}}^{3d}|}$ as the set of nodes and $\hat{\mathcal{E}}^{3d} = \{\hat{e}_{ij}\}_{i,j \in [1, |\hat{\mathcal{V}}^{3d}|]}$ as the set

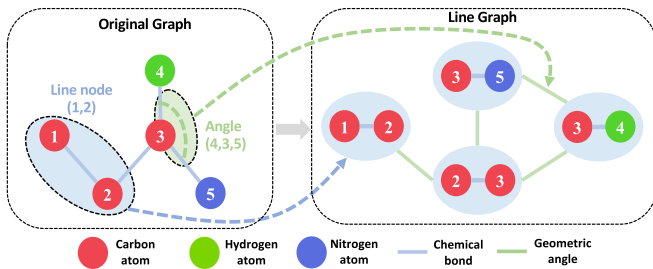


Fig. 2. Illustration of transforming a molecular graph to its line graph. This transformation preserves the inherent adjacency information in the original graph while emphasizing structural information at finer granularity levels.

of edges in $\hat{\mathcal{G}}^{3d}$. Fig. 2 illustrates the process of line graph transformation.

B. Dual-Modality Line Graph Transformers

Although transformer architecture has demonstrated remarkable performance in various tasks involving sequence data, its direct application to graph-structured data still remains challenging. The primary obstacle lies in the fact that the self-attention and feedforward network modules are order-agnostic to the input features, and the classic positional encoding [55] fails to capture graph structural information. To handle this problem, we develop dual-modality line graph transformers, which introduce efficient structural encoding methods to incorporate the topological and geometric structures, respectively. The detailed architecture is shown in Fig. 3.

1) *2-D Line Graph Transformer*: We introduce three distinct structural encoding methods: eigenvector positional encoding, path length encoding, and path node encoding, thus integrating the topological information of 2-D line graphs into the transformer architecture. These encoding methods serve as the inductive bias to convey structural information, leading to more robust and discriminative graph representations.

a) *Eigenvector positional encoding*: For the topological line graph, we use the eigenvectors of its normalized Laplacian matrix as positional encoding. Laplacian matrix is defined as the difference between the degree matrix and the adjacency matrix, representing the topology of a graph. Furthermore, Laplacian eigenvectors can distinguish the local position information of different nodes while preserving the global topology structure [60], [61]. For each node in the topology line graph, its corresponding eigenvector is viewed as a node positional encoding and added to the node feature for position-aware feature input

$$\mathbf{x}_i^{2d} = \mathbf{n}_i^{2d} + \mathbf{W}_v \mathbf{v}_i^{2d} \quad (10)$$

where $\mathbf{v}_i^{2d} \in \mathbb{R}^K$ is the K smallest nontrivial eigenvectors for the i th node in the topological line graph, and $\mathbf{W}_v \in \mathbb{R}^{\Theta_v \times K}$ denotes a learnable projection matrix. The eigenvector positional encoding for graph-structured data can be regarded as a natural generalization of the sine-cosine positional encoding for sequence-structured data. By adding it to the input node features, the transformer can effectively capture their positional information in the graph.

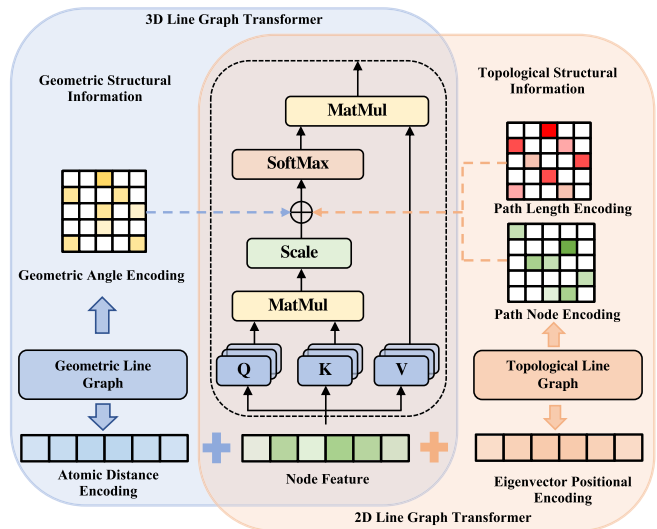


Fig. 3. Illustration of the dual-modality line graph transformer architecture. The blue part is the 3-D line graph transformer capturing the geometric structural information, and the orange part is the 2-D line graph transformer incorporating the topological structural information. Both of them are derived from the vanilla transformer backbone.

b) *Path length encoding*: Different from general structured data in Euclidean space, e.g., texts and images, graphs do not have a canonical order and can only lie in a non-Euclidean space connected by edges. To model full structural information, the spatial relations between nodes should be measured. Following the previous studies [13], [34], we derive the shortest path between each pair of nodes, and then leverage the path length and the node features along the path to capture spatial information. For the path length encoding, we assign a learnable embedding vector for each length scalar, then project it as a bias term in the self-attention module. Given a node pair $(\hat{v}_i, \hat{v}_j)$ in the topological line graph, the path length encoding can be written as

$$\begin{aligned} \mathbf{p}_{ij} &= \text{SPL}(\hat{v}_i, \hat{v}_j) \\ b_{ij} &= \mathbf{p}_{ij}^T \mathbf{w}_p \end{aligned} \quad (11)$$

where $\text{SPL}(\cdot)$ denotes an embedding function that determines a learnable embedding vector $\mathbf{p}_{ij} \in \mathbb{R}^p$ indexed by the length of the shortest path between node $\hat{v}_i$ and $\hat{v}_j$, and $\mathbf{w}_p \in \mathbb{R}^p$ is a learnable projection vector.

c) *Path node encoding*: As the different nodes along the shortest path can distinguish spatial relations in graphs, we encode the path node features as additional structural information into the model. For each node pair $(\hat{v}_i, \hat{v}_j)$, we compute an average bias term of the dot products of the node feature and a learnable projection vector along the path, which is defined as

$$c_{ij} = \frac{1}{N_{ij}} \sum_{n=1}^{N_{ij}} (\mathbf{x}_n^{2d})^T \mathbf{w}_n \quad (12)$$

where N_{ij} is the shortest path length between node $\hat{v}_i$ and $\hat{v}_j$, $\mathbf{x}_n^{2d}$ is the n th 2-D node feature along the path, and $\mathbf{w}_n \in \mathbb{R}^{\Theta_v}$ denotes the n th learnable projection vector indexed by the path position.

d) *Topological bias in 2-D attention*: We incorporate the proposed three structural encodings into the transformer architecture, which captures full structural information in the topological line graph. Specifically, the eigenvector positional encoding is added to the original node feature as the input feature. Path length and path node encodings are two structural bias terms in the self-attention module. We can compute each element (i, j) in the attention matrix as follows:

$$\mathbf{A}_{ij}^{2d} = \frac{(\mathbf{W}_q^{2d} \mathbf{x}_i^{2d})^T (\mathbf{W}_k^{2d} \mathbf{x}_j^{2d})}{\sqrt{d_k}} + b_{ij} + c_{ij} \quad (13)$$

where $\mathbf{W}_q^{2d} \in \mathbb{R}^{d_k \times \Theta_v}$ and $\mathbf{W}_k^{2d} \in \mathbb{R}^{d_k \times \Theta_v}$ are the learnable projection matrices. Similar to the vanilla transformer, we augment the self-attention module by extending it to the multihead mechanism. Note that the structural bias terms b_{ij} and c_{ij} should also be computed by separate projection vectors within different heads. Then, we adapt the feedforward network with layer normalization [56] and skip connection operations [57] over the output of the self-attention module.

2) *3-D Line Graph Transformer*: To incorporate the geometric graph information into the transformer architecture, we consider two novel structural encoding methods: atomic distance encoding and geometric angle encoding. By leveraging the constructed geometric line graph, the atomic distance and geometric angle can be embedded into the node and edge representations in a natural manner. For the atomic distance encoding, we concatenate the atomic feature embedding and pairwise distance embedding in (8), which provides a relative position encoding as the input feature to the following modeling, that is, $\mathbf{x}_i^{3d} = \mathbf{h}_{uv}^{3d}$.

a) *Geometric angle encoding*: As an invariant spatial feature, the geometric angle between the adjacent bonds plays an important role in determining the spatial structure of 3-D molecules. Similarly, we derive the shortest path between any two nodes in the geometric line graph. Since the angle embedding is viewed as the line edge feature by (9), we can encode the edge features along the path as the structural information. Given a node pair $(\hat{v}_i, \hat{v}_j)$, an average bias term is computed by taking the dot product of the edge feature and a learnable projection vector along the shortest path

$$g_{ij} = \frac{1}{N_{ij} - 1} \sum_{m=1}^{N_{ij}-1} (\mathbf{e}_m^{3d})^T \mathbf{w}_m \quad (14)$$

where N_{ij} is the shortest path length, $\mathbf{e}_m^{3d}$ is the m th 3-D edge feature along the path, and $\mathbf{w}_m \in \mathbb{R}^{\Theta_e}$ denotes the m th learnable projection vector indexed by the path position.

b) *Geometric bias in 3-D attention*: We introduce the pairwise distance and angle encodings into the transformer architecture, capturing the geometric structural information. For each element (i, j) in the attention matrix of the 3-D line graph transformer, we have

$$\mathbf{A}_{ij}^{3d} = \frac{(\mathbf{W}_q^{3d} \mathbf{x}_i^{3d})^T (\mathbf{W}_k^{3d} \mathbf{x}_j^{3d})}{\sqrt{d_k}} + g_{ij} \quad (15)$$

where $\mathbf{W}_q^{3d} \in \mathbb{R}^{d_k \times \Theta_v}$ and $\mathbf{W}_k^{3d} \in \mathbb{R}^{d_k \times \Theta_v}$ are learnable projection matrices. As with the 2-D line graph transformer,

we leverage the multihead attention mechanism to enhance the self-attention module, then feed the output into a feedforward network with layer normalization and skip connection operations.

c) *Discussion*: Here, we discuss two advantages of the designed dual-modality architecture as follows.

- 1) Both 2-D and 3-D line graph transformers are derived from the vanilla transformer backbone but encode different topological and geometric information, respectively. The consistent backbone across two modalities ensures that the encoded representations are directly comparable, and the knowledge can be transferred in the subsequent intermodality learning.
- 2) The structural information is preserved in the node and edge features by transforming original molecular graphs into line graphs. This transformation ensures that graph structures can be more effectively captured by most graph-based machine learning methods.

Despite different terminologies, the concept of molecular line graph has already been applied in previous studies. However, they separate the line graph into atom-bond and bond-angle graphs [21], [58], or consider the transformation only on 2-D molecules [13].

C. Pretraining Task Construction

The efficiency of the pretraining framework significantly depends on the construction of pretraining tasks. In this work, we design two complementary tasks from both intramodality and intermodality levels: masked line node prediction and dual-modality contrastive learning.

1) *Masked Line Node Prediction*: For the pretraining at the intramodality level, we leverage node masking to let the model learn the regularities of the atom/bond attributes. Following the “masked language model (MLM)” training objective in BERT [62], we randomly sample a proportion of line nodes in the topological and geometric line graphs, and replace them with the masking procedures: 1) masking 80% of the selected nodes using a special *MASK* indicator; 2) replacing 10% of the selected nodes with other random nodes; and 3) keeping the remaining 10% nodes unchanged. Then, we apply 2-D/3-D line graph transformers to obtain the corresponding line node embeddings. Finally, a multilayer perceptron (MLP) is used on top of the masked embeddings to predict the types of the original nodes with a cross-entropy loss. The mask losses on the 2-D and 3-D modalities are denoted as $\mathcal{L}_{mask}^{2d}$ and $\mathcal{L}_{mask}^{3d}$, respectively.

2) *Dual-Modality Contrastive Learning*: As the 2-D and 3-D modalities of the same molecule are correlated and can provide complementary knowledge to each other, we can enhance molecular SSL by designing an intermodality pretraining task. Specifically, we leverage contrastive learning to maximize the mutual information between the 2-D and 3-D representations of a molecule. In each line graph, we create a virtual node connected to all other nodes. Its output represents the graph-level embedding and can be used for intermodality contrast. For each molecule, we first extract the graph-level embeddings from the masked topological and geometric line

TABLE I
STATISTICS OF MOLECULAR PROPERTY DATASETS

Dataset	# Molecules	# Tasks	Source	Type
BBBP	2,039	1	MoleculeNet	Classification
Sider	1,427	27	MoleculeNet	Classification
ClinTox	1,478	2	MoleculeNet	Classification
BACE	1,513	1	MoleculeNet	Classification
Tox21	7,831	12	MoleculeNet	Classification
MUV	93,087	17	MoleculeNet	Classification
HIV	41,127	1	MoleculeNet	Classification
Estrogen	3,122	2	ChEMBL	Classification
MetStab	2,267	2	MetStabOn	Classification
ESOL	1,128	1	MoleculeNet	Regression
Freesolv	642	1	MoleculeNet	Regression
Lipo	4,200	1	MoleculeNet	Regression
Malaria	9,999	1	Antimalarial	Regression
CEP	29,978	1	Clean Energy Project	Regression
QM9	132,481	8	MoleculeNet	Regression

graphs, and then adapt the individual projection heads to map them into a common representation space, resulting in the derivation of $\mathbf{z}_i^{2d}$ and $\mathbf{z}_i^{3d}$. The 2-D–3-D latent vectors from the same molecule are regarded as positive pairs and negative pairs otherwise. Based on the InfoNCE loss [63], our dual-modality contrastive learning is written as

$$\mathcal{L}_{cl}^{2d} = - \sum_{i=1}^N \log \frac{\exp(\text{sim}(\mathbf{z}_i^{2d}, \mathbf{z}_i^{3d}) / \tau)}{\sum_{j=1}^M \exp(\text{sim}(\mathbf{z}_i^{2d}, \mathbf{z}_j^{3d}) / \tau)} \quad (16)$$

$$\mathcal{L}_{cl}^{3d} = - \sum_{i=1}^N \log \frac{\exp(\text{sim}(\mathbf{z}_i^{3d}, \mathbf{z}_i^{2d}) / \tau)}{\sum_{j=1}^M \exp(\text{sim}(\mathbf{z}_i^{3d}, \mathbf{z}_j^{2d}) / \tau)} \quad (17)$$

$$\mathcal{L}_{cl} = \frac{1}{N} (\mathcal{L}_{cl}^{2d} + \mathcal{L}_{cl}^{3d}) \quad (18)$$

where $\mathbf{z}_i^{2d}$ and $\mathbf{z}_i^{3d}$ denotes two latent vectors in 2-D and 3-D modalities from the same molecule, τ is a temperature coefficient, $\text{sim}(\cdot)$ measures the dot similarity between two vectors, M is the batch size, and N is the number of molecules in the dataset. Dual-modality contrastive learning can simultaneously align the positive pairs of the same molecule and distinguish them from negative pairs, thereby enabling the 2-D/3-D representation to extract complementary information from its 3-D/2-D counterpart.

Finally, we combine the intramodality mask losses and intermodality contrastive loss into an overall loss function as follows:

$$\mathcal{L}_{obj} = \lambda_1 (\mathcal{L}_{mask}^{2d} + \mathcal{L}_{mask}^{3d}) + \lambda_2 \mathcal{L}_{cl} \quad (19)$$

where λ_1 and λ_2 are weighting coefficients. The training objective is to minimize the overall loss $\mathcal{L}_{obj}$. As a result, the mask generative losses capture useful semantics of molecules in two modalities, while the contrastive loss can simultaneously provide complementary information from each other. Consequently, our SSL framework can produce a robust and discriminative molecular representation that encodes both topological and geometric information.

V. EXPERIMENTS

In this section, we first introduce the experimental settings, including benchmark datasets, implementation details, and

baselines. Then, the pretrained models are fine-tuned to various downstream tasks for evaluation.

A. Experimental Settings

1) *Benchmark Datasets*: For pretraining, we use unlabeled GEOM dataset [64], which comprises 304k molecules with precise 2-D and 3-D structures. Most molecules have multiple 3-D conformers, but the conformer with the lowest energy is deemed the most stable and has the highest probability. Therefore, we take the lowest-energy conformer of each molecule as its 3-D structure.

For downstream fine-tuning, to comprehensively evaluate model performance, we leverage 15 labeled molecular property datasets from MoleculeNet [65], and other public sources [66], [67], [68], [69] as downstream prediction tasks. These sources include nine classification datasets and six regression datasets. These benchmark datasets have been widely used in previous studies [9], [13], [18], [25] and provide comprehensive coverage of molecular properties across different domains, such as physiology, biophysics, materials science, and quantum chemistry. The detailed statistics of these datasets are summarized in Table I.

2) *Data Splitting*: In machine learning, datasets are typically divided into training, validation, and test sets using a random split. However, this split strategy does not consider the structural similarities between molecules. As suggested in MoleculeNet [65], we adopt a scaffold split to create a more challenging yet realistic evaluation for molecular property prediction. Each downstream dataset is divided into training, validation, and test sets with an 8:1:1 ratio. We perform three independent scaffold splits using different random seeds and report the mean and standard deviation of the evaluation metrics.

3) *Baselines*: We comprehensively evaluate the performance of Galformer against ten state-of-the-art SSL methods on molecules. These methods include GraphCL [18] and JOAO [16], which are 2-D contrastive methods leveraging data augmentation on 2-D molecular graphs to facilitate representation learning. Mix-Key [50] is a data augmentation method based on the key structures of 2-D molecules, and MTSSMol [46] employs multitask learning, integrating two pretraining strategies and an unsupervised learning method on 2-D representations to enhance the prediction performance. GROVER [10] and KPGT [13] are 2-D generative methods that incorporate domain knowledge for the design of SSL tasks. DMP [70] combines the 1-D sequence and 2-D graph of a molecule to learn dual-modality representations. In addition, 3DInformax [22], GraphMVP [25], and 3D-PGT [54] are 2-D–3-D pretraining methods with GNN-based backbones. 3DInformax aims to maximize the mutual information between 2-D and 3-D representations via contrastive learning, while GraphMVP adopts a hybrid strategy to learn complementary 2-D–3-D information. 3D-PGT proposes three pretext tasks and introduces an energy-based metric to integrate them to leverage the 3-D molecule data in pretraining.

4) *Graph Representation of Molecules*: The 2-D and 3-D molecular graphs capture high-level topological and geometric relations, which are significant to the functionality of molecules. For each atom in a given molecule, we create

TABLE II
GALFORMER HYPERPARAMETER CONFIGURATION

Hyperparameter	Pre-training	Fine-tuning
Training epoch	50	50
Batch size	256	32
Peak learning rate	$2e^{-4}$	$[3e^{-5}, 1e^{-5}, 3e^{-6}]$
Gradient clipping	5.0	5.0
Activation function	ReLU	ReLU
Weight decay	$1e^{-6}$	$[0, 1e^{-6}]$
Dropout rate	0.1	0.1
Number of backbone layer	12	12
Hidden size	768	768
Attention head	12	12
Distance Gaussian Basis kernel	256	256
Angle Gaussian Basis kernel	128	128
Line node mask rate	0.4	NA
Temperature coefficient τ	0.1	NA
Multi-task coefficient λ_1, λ_2	1.0, 1.0	NA

a 101-D one-hot vector representing the atom type and a 36-D integer vector capturing atom properties. These atom properties include formal charge, number of hydrogen atoms, presence of a chiral center, chirality type, number of radical electrons, hybridization state, bond degree, atomic mass, and aromaticity. We concatenate the atom type vector and atom property vector to create the initial node feature in a 2-D topological graph. Meanwhile, in a 3-D geometric graph, we utilize the individual atom-type vector as the initial node feature.

For each bond in a given molecule, we generate a 5-D one-hot vector to encode the bond type and a 9-D integer vector capturing bond properties, including ring membership, stereochemistry, and conjugation status. These vectors are concatenated to form the initial edge feature in a 2-D topological graph. In the 3-D geometric graph, we calculate the pairwise atomic distance and represent it as a vector using a Gaussian basis kernel function, which serves as the initial edge feature.

5) Implementation Details: We implement Galformer mainly in PyTorch [72] and DGL [73], along with useful functions from RDKit [74], DGLLifeSci [75], and Scikit-learn [76]. The hyperparameter settings for pretraining and fine-tuning are detailed in Table II. For training optimization, we use the Adam optimizer [77] with a polynomial decay learning rate scheduler. Galformer is built with 12-layer 2-D and 3-D line graph transformers to encode dual-modality molecular information, using a hidden size of 768 and 12 attention heads. The batch size is set to 256, and the model is pretrained for 50 epochs. The peak learning rate is set to $2e^{-4}$, with an Adam weight decay of $1e^{-6}$. In addition, we use a mask ratio of 0.4 and set the temperature coefficient τ to 0.1. For balanced optimization, the multitask weighting coefficients λ_1 and λ_2 are all set to 1. After pretraining, we concatenate the graph-level embedding and the averaged line node embedding as the molecular representation for downstream tasks.

During downstream task fine-tuning, we use a two-layer MLP with ReLU activation as the prediction head, which is added on top of the pretrained 2-D backbone network. For classification, binary cross-entropy loss is used, while root-

mean-square error (RMSE) loss is used for regression tasks. The hidden dimension of the prediction head is set to 256, and the batch size is set to 32. We perform a grid search for hyperparameters, exploring six combinations of learning rates $[3e^{-5}, 1e^{-5}, 3e^{-6}]$ and weight decays $[0, 1e^{-6}]$ to identify the best configuration for each task. The model is fine-tuned for 50 epochs, and the best-performing epoch on the validation set is used to report the testing score. All experiments are conducted on Tesla A100 GPUs.

6) Evaluation Metrics: We use two metrics to evaluate the performance of downstream tasks: the area under the receiver operating characteristic curve (AUROC) for classification tasks, and RMSE and mean absolute error (MAE) for regression tasks. For classification datasets with multibinary labels, we calculate the average AUROC across all tasks as the final metric.

B. Downstream Task Evaluation

Tables III and IV present the testing performance of all methods on classification and regression tasks, respectively. The results provide the following observations.

- 1) Galformer consistently performs the overall best on classification and regression tasks, demonstrating its effectiveness. Compared to previous SOTA results from baselines, Galformer has an overall relative improvement of 0.013 on the classification tasks and 0.018 on the regression tasks.
- 2) Among contrastive learning baselines, GraphMVP, 3DInformax, and 3D-PGT, which leverage both 2-D and 3-D graphs in pretraining, perform better on the whole average than GraphCL and JOAO, which only use 2-D graphs. This indicates that incorporating 3-D geometry can effectively facilitate self-supervised molecular learning. Furthermore, Galformer achieves a more significant improvement compared with the 2-D-3-D contrastive baselines. This result can be attributed to our complementary pretraining tasks, which can adaptively capture both intermodality and intramodality information.
- 3) KPGT, DMP, and GROVER outperform other baselines for most tasks. One potential explanation is that they both employ transformer-based architectures as backbone networks, leading to higher model capacity and more expressive power than message-passing models. Nevertheless, the lack of leveraging 3-D geometry limits their ability to fully learn molecular representations.
- 4) MTSSMol clearly outperforms other methods on toxicity-centered tasks, ClinTox and BBBP, which can be attributed to a pretraining strategy tailored to capture toxicity-related structural patterns. In contrast, Galformer consistently achieves improved performance than MTSSMol on all other downstream tasks, demonstrating its effectiveness across a broader range of molecular properties. Overall, Galformer exhibits a notable capability to surpass the inherent limitations in previous methods, consequently achieving more generalizable molecular property prediction.

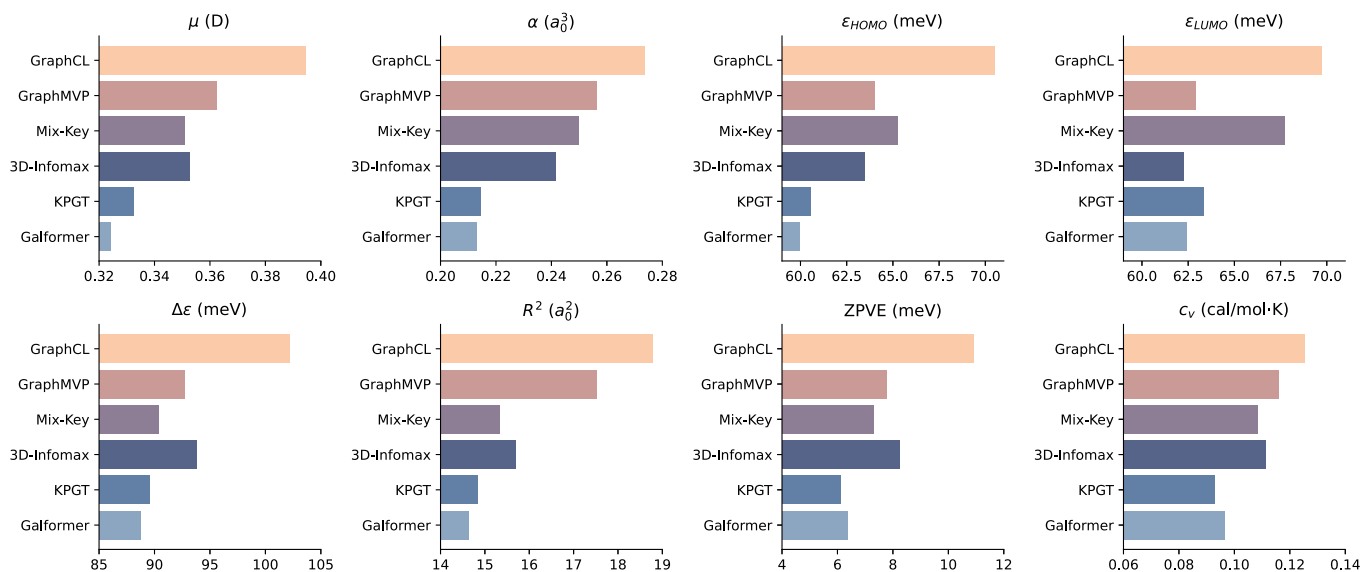


Fig. 4. Regression MAE for eight standard prediction tasks on the QM9 dataset [65] shows that Galformer outperforms the other five baselines on the whole. A lower value indicates better performance in all eight tasks.

Fig. 4 shows the performance comparison on eight standard quantum chemical property prediction tasks from the QM9 dataset [65]. Here, we selected five representative baselines for comparison based on reproducibility and relevance. The results show that Galformer achieves the best overall performance, highlighting its effectiveness in predicting quantum chemical properties.

Although Galformer achieves the highest overall average performance across both classification and regression benchmarks, it does not achieve the best performance on every individual task. This situation can be attributed to two main factors. First, Galformer is designed as a unified and generalizable molecular pretraining framework rather than a model specialized for particular downstream objectives. Its primary goal is to learn transferable and broadly applicable molecular representations. This design slightly limits task-specific gains but enhances robustness and consistency across diverse properties. Second, Galformer is pretrained on the GEOM dataset, which contains approximately 304k molecules. This scale is significantly smaller than the pretraining sets used by some baselines, such as KPGT with 2 million molecules and MTSSMol with 10 million molecules. The smaller scale of pretraining data can lead to suboptimal performance on certain downstream tasks. Nevertheless, Galformer is expected to benefit from the future expansion of large-scale paired 2-D–3-D molecular datasets.

C. Ablation Studies

We conduct ablation studies to analyze the effectiveness of several critical designs in our proposed framework.

1) *Impact of Different Temperature Coefficients:* For the dual-modality contrastive learning, we set the temperature coefficient (τ) to control the smoothness of the softmax distribution over similarity scores. Fig. 5 illustrates the effect of varying τ values on the contrastive loss curves during

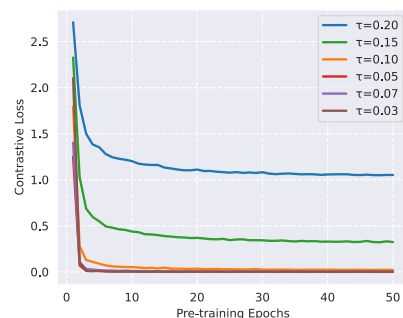


Fig. 5. Pretraining contrastive loss of Galformer under different temperature coefficients (τ).



Fig. 6. AUROC (%) performance of Galformer on four downstream tasks, under different temperature coefficients τ (best in bold).

pretraining. The larger τ values (0.15 and 0.20) lead to higher and more slowly converging losses, while smaller τ values result in faster convergence and lower final loss values. This indicates that a lower temperature makes the similarity distribution more discriminative, enhancing the effectiveness of contrastive learning. Fig. 6 indicates the performance variations on the four downstream datasets. The heatmap shows

TABLE III

TEST GALFORMER PERFORMANCE IN AUROC AGAINST TEN EXISTING METHODS ON NINE DOWNSTREAM CLASSIFICATION DATASETS (BEST IN BOLD, SECOND BEST IN UNDERLINE). A LARGER VALUE INDICATES BETTER PERFORMANCE (MARKED BY $\uparrow$). EACH EXPERIMENT IS CONDUCTED INDEPENDENTLY THREE TIMES WITH DIFFERENT RANDOM SEEDS FOR SCAFFOLD SPLIT

Method\Dataset	Classification accuracy (AUROC) $\uparrow$									
	BBBP	Sider	ClinTox	BACE	Tox21	MUV	HIV	Estrogen	MetStab	Avg.
GraphCL [18]	0.901 _(0.033)	0.625 _(0.023)	0.642 _(0.120)	0.868 _(0.025)	0.823 _(0.015)	0.762 _(0.027)	0.776 _(0.007)	0.891 _(0.020)	0.794 _(0.037)	0.787
JOAO [16]	0.900 _(0.031)	0.629 _(0.020)	0.707 _(0.068)	0.866 _(0.035)	0.825 _(0.011)	0.766 _(0.048)	0.769 _(0.004)	0.869 _(0.050)	0.814 _(0.031)	0.794
GROVER [10]	0.923 _(0.029)	0.645 _(0.010)	0.894 _(0.032)	0.882 _(0.030)	0.840 _(0.016)	0.831 _(0.035)	0.773 _(0.014)	0.903 _(0.030)	0.822 _(0.039)	0.835
3DInfomax [22]	0.905 _(0.033)	0.634 _(0.029)	0.724 _(0.104)	0.862 _(0.023)	0.819 _(0.021)	0.803 _(0.007)	0.750 _(0.010)	0.871 _(0.039)	0.811 _(0.043)	0.798
GraphMVP [25]	0.918 _(0.019)	0.652 _(0.027)	0.705 _(0.092)	0.866 _(0.028)	0.832 _(0.017)	0.787 _(0.039)	0.791 _(0.004)	0.892 _(0.033)	0.827 _(0.043)	0.808
3D-PGT [54]	0.894 _(0.027)	0.634 _(0.012)	0.732 _(0.109)	0.840 _(0.011)	0.812 _(0.019)	0.763 _(0.034)	0.770 _(0.015)	0.796 _(0.123)	0.888 _(0.029)	0.792
MTSSMol [46]	0.949 _(0.026)	0.596 _(0.023)	0.981 _(0.022)	0.848 _(0.005)	0.779 _(0.011)	0.737 _(0.003)	0.764 _(0.018)	0.858 _(0.013)	0.803 _(0.051)	0.813
Mix-Key [50]	0.915 _(0.018)	0.633 _(0.025)	0.765 _(0.043)	0.876 _(0.025)	0.818 _(0.025)	0.773 _(0.028)	0.770 _(0.048)	0.896 _(0.005)	0.826 _(0.040)	0.808
DMP [70]	0.930 _(0.031)	0.655 _(0.025)	0.908 _(0.032)	0.890 _(0.033)	0.830 _(0.022)	0.835 _(0.020)	0.798 _(0.015)	0.910 _(0.025)	0.853 _(0.046)	0.845
KPGT [13], [71]	0.927 _(0.028)	<u>0.658</u> _(0.013)	<u>0.915</u> _(0.027)	<u>0.893</u> _(0.032)	<u>0.847</u> _(0.013)	0.829 _(0.047)	0.768 _(0.007)	<u>0.915</u> _(0.028)	0.846 _(0.052)	0.844
Galformer	<u>0.933</u> _(0.027)	0.681 _(0.011)	0.910 _(0.030)	0.897 _(0.026)	0.852 _(0.015)	0.841 _(0.023)	<u>0.796</u> _(0.017)	0.936 _(0.010)	<u>0.875</u> _(0.029)	0.858

TABLE IV

TEST GALFORMER PERFORMANCE IN RMSE AGAINST TEN EXISTING METHODS ON FIVE DOWNSTREAM REGRESSION DATASETS (BEST IN BOLD, SECOND BEST IN UNDERLINE). A LOWER VALUE INDICATES BETTER PERFORMANCE (MARKED BY $\downarrow$). EACH EXPERIMENT IS CONDUCTED INDEPENDENTLY THREE TIMES WITH DIFFERENT RANDOM SEEDS FOR SCAFFOLD SPLIT

Method\Dataset	Regression error (RMSE) $\downarrow$					Avg.
	ESOL	Freesolv	Lipo	Malaria	CEP	
GraphCL [18]	1.253 _(0.023)	2.216 _(0.117)	0.762 _(0.029)	1.115 _(0.127)	1.251 _(0.029)	1.319
JOAO [16]	1.203 _(0.061)	2.010 _(0.189)	0.759 _(0.015)	1.117 _(0.122)	1.278 _(0.021)	1.273
GROVER [10]	0.928 _(0.106)	1.998 _(0.280)	0.703 _(0.023)	1.092 _(0.086)	1.064 _(0.085)	1.157
3DInfomax [22]	1.167 _(0.148)	2.275 _(0.391)	0.755 _(0.025)	1.075 _(0.103)	1.277 _(0.019)	1.310
GraphMVP [25]	0.995 _(0.113)	1.806 _(0.164)	0.736 _(0.051)	1.101 _(0.107)	1.248 _(0.010)	1.177
3D-PGT [54]	0.830 _(0.067)	1.505 _(0.042)	0.794 _(0.036)	1.136 _(0.131)	0.976 _(0.050)	1.048
MTSSMol [46]	1.232 _(0.144)	2.141 _(0.443)	0.824 _(0.052)	1.141 _(0.117)	1.149 _(0.079)	1.297
Mix-Key [50]	1.004 _(0.026)	1.820 _(0.427)	0.753 _(0.033)	0.899 _(0.118)	1.075 _(0.035)	1.110
DMP [70]	0.815 _(0.048)	1.335 _(0.251)	0.658 _(0.020)	1.095 _(0.085)	1.128 _(0.042)	1.006
KPGT [13], [71]	<u>0.802</u> _(0.096)	<u>1.230</u> _(0.231)	0.621 _(0.018)	1.083 _(0.082)	1.020 _(0.100)	<u>0.951</u>
Galformer	0.741 _(0.087)	1.213 _(0.200)	<u>0.628</u> _(0.015)	<u>1.066</u> _(0.092)	<u>1.015</u> _(0.084)	0.933

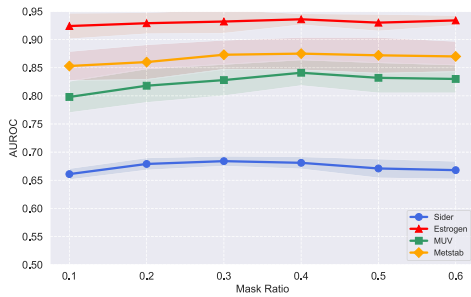


Fig. 7. AUROC performance of Galformer on four downstream tasks, under different mask ratios.

that $\tau = 0.10$ consistently achieves strong and balanced performance across tasks, including the best AUROC on Metstab, Estrogen, and HIV. Although smaller τ values lead to lower pretraining loss, they can limit downstream generalization due to reduced representation diversity and a less adaptable embedding space. A temperature coefficient of 0.1 provides a good tradeoff between training stability and downstream performance.

2) *Impact of Different Mask Ratios*: For the masked line node prediction, we randomly mask a proportion of nodes in both the 2-D and 3-D line graphs. Fig. 7 shows the impact of

different mask ratios on four benchmark datasets. The results indicate that Galformer achieves optimal performance with a mask ratio of 40%. This ratio is significantly higher than those typically used in previous works [9], [12], [62], which generally employed ratios below 25%. The additional contrastive learning provides complementary knowledge from different modalities, allowing Galformer to handle more challenging tasks with a higher mask ratio. This shares similarity with the success of masked autoencoders (MAEs) [78] in the vision domain, where a higher mask ratio can better capture long-range dependencies and improve model generalization.

3) *Impact of Pretraining Strategy and Backbone Network*: To verify the effectiveness of our pretraining strategy and backbone network, we study Galformer without pretraining and replace the 2-D line graph transformer backbone with Galformer without line node, vanilla transformer [55], and Graphormer [34]. Table V shows the performance comparison, where Galformer achieves significant improvement after pretraining. Specifically, our pretraining strategy yields an average improvement of 0.048 in AUROC and 0.115 in RMSE. Compared to the other pretrained variants with different backbones, Galformer still improves up to 0.066 in AUROC and 0.262 in RMSE. This improvement can be attributed to our well-designed graph structural encodings in the 2-D

TABLE V

ABLATION STUDY ON THE AVERAGED AUROC FOR CLASSIFICATION DATASETS AND RMSE FOR REGRESSION DATASETS (EXCLUDING QM9), EVALUATING THE ABSENCE OF PRETRAINING AND DIFFERENT BACKBONES (BEST IN BOLD, SECOND BEST IN UNDERLINE)

Backbone	Classification Avg. AUROC $\uparrow$	Regression Avg. RMSE $\downarrow$
No Pre-training	0.810	1.048
No Line Node	0.792	1.195
Vanilla Transformer [55]	0.825	1.012
Graphormer [34]	<u>0.854</u>	<u>0.951</u>
Galformer	0.858	0.933

TABLE VI

ABLATION STUDY ON THE AVERAGED AUROC FOR CLASSIFICATION DATASETS AND RMSE FOR REGRESSION DATASETS (EXCLUDING QM9), EVALUATING DIFFERENT CONTRASTIVE LOSSES (BEST IN BOLD, SECOND BEST IN UNDERLINE)

Contrastive Loss (CL)	Classification Avg. AUROC $\uparrow$	Regression Avg. RMSE $\downarrow$
w/o Mask Loss	0.815	1.020
w/o Contrastive Loss	0.838	0.986
NT-Xent [79]	0.850	<u>0.954</u>
EMB-NCE [25]	<u>0.852</u>	0.956
InfoNCE [63]	0.858	0.933

line graph transformer, effectively capturing key molecular structural information.

4) *Impact of Different Contrastive Losses*: To enhance the model’s understanding of molecules at the intramodality level, Galformer employs an intramodality masked line node prediction task in pretraining. To demonstrate the effectiveness of our mask-and-prediction task, we perform an ablation study. Shown as “w/o Mask Loss” in Table VI, Galformer achieves an improvement of 0.043 in AUROC and 0.087 in RMSE. This observation highlights the necessity of the masked line node prediction task in pretraining.

For modeling intermodality relations, Galformer leverages contrastive learning to bridge the modality gap between 2-D and 3-D representations. To validate the effectiveness of our equipped InfoNCE loss, we implement three model variants: one without contrastive learning, and two others with NT-Xent [79] and EMB-NCE [25] contrastive losses, respectively. As shown in Table VI, contrastive loss can improve performance on both classification and regression tasks, indicating that incorporating 3-D information during pretraining can significantly enhance model performance toward downstream tasks. Furthermore, InfoNCE loss consistently outperforms the other variants, which verifies its effectiveness in cross-modality contrastive learning.

5) *Impact of Data Scale in Pretraining*: To evaluate the scalability of Galformer as a pretraining pipeline, we use 10%, 30%, 50%, 70%, 80%, and 100% of the pretraining dataset and perform 14 downstream tasks in Tables III and IV. The results in Fig. 8 indicate that Galformer’s performance linearly increases as more pretraining data are included, demonstrating Galformer’s superior scalability as a pretraining model and its feasibility for large-scale molecular pretraining.

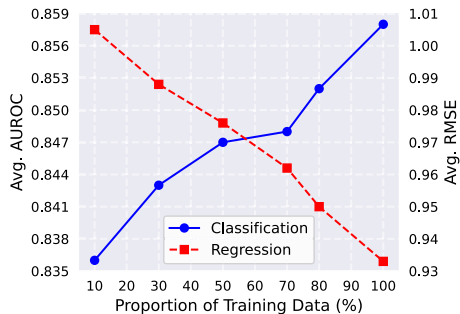


Fig. 8. Performance of Galformer as the pretraining data scale increases. Avg. AUROC and Avg. RMSE are the averages over nine classification datasets and five regression datasets (Avg. in Tables III and IV). The results demonstrate the scalability of Galformer as a pretraining method.

D. Molecular Representation Visualization

To intuitively investigate the impact of pretrained molecular representations without fine-tuning, we randomly select 800 molecules from the BBBP and ESOL datasets, respectively, and utilize *t*-SNE [80] to project these representations onto a 2-D embedding space for visualization. In the BBBP dataset, each molecule is associated with a binary class label that indicates its barrier permeability property. In the ESOL dataset, the original label is a continuous variable that denotes the water solubility of the molecule. We use the median value of 0.057 mols/L in ESOL as a threshold: molecules with solubility above this threshold are considered positive. Otherwise, they are negative. Fig. 9 shows the visualization results, using the Davies–Bouldin (DB) index [81] as a metric to quantify cluster separation. The lower the DB index, the better the separation. The results indicate that both pretrained GraphMVP and Galformer achieve better separation of labeled molecules than models without pretraining. In particular, Galformer further decreases the DB index to 1.96 on BBBP, compared to 3.88 on GraphMVP. Importantly, these results are obtained in a zero-shot learning setting without fine-tuning, showing that Galformer effectively incorporates generalizable domain knowledge during pretraining and thus significantly improves the quality of the learned molecular representations.

E. Attention Weight Visualization

In Galformer, a virtual node is fully connected to all realistic line nodes, and its embedding is used for downstream prediction tasks. To estimate the contributions of different substructures, we extract the attention weights directed toward the virtual node from the last layer of the line graph transformer. The line nodes with the top 20% of attention weights are then selected and highlighted as the key substructures.

As shown in Fig. 10, we visualize four molecules from the BBBP dataset to examine whether Galformer emphasizes chemically meaningful substructures. For the permeable molecules Trifluoperidol and Fletazepam, the highlighted nodes concentrate around the trifluoromethyl group, a group known to increase lipophilicity and facilitate membrane permeability [82]. This indicates that the model tends to focus on

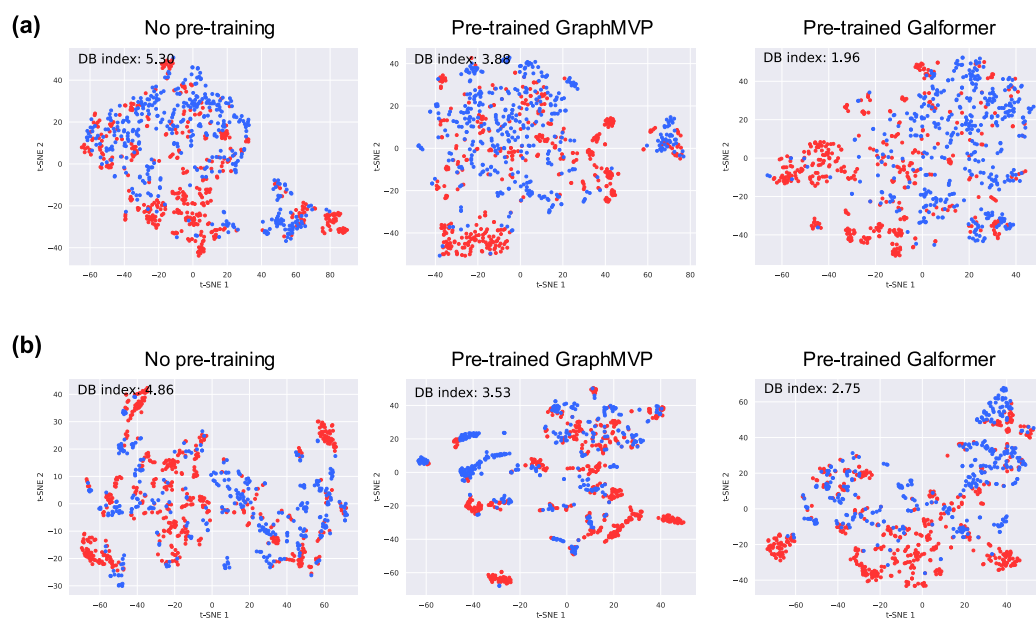


Fig. 9. Visualization of the pretrained molecular representations on the BBBP and ESOL datasets via t-SNE. All results are without fine-tuning by ground-truth labels. (a) BBBP dataset: color represents binary labels for the barrier permeability property. (b) ESOL dataset: color represents binary labels for water solubility. DB index measures the degree of cluster separation. The lower the DB index, the better the separation.

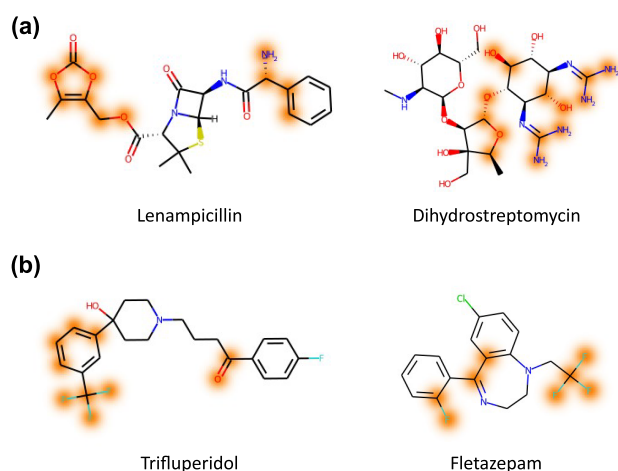


Fig. 10. Visualization of molecules from BBBP, with high-attention substructures highlighted in orange. (a) BBB-impermeable molecules (label 0), Lenampicillin and Dihydrostreptomycin, attention weights focus on highly polar regions (e.g., carboxylic acid and guanidine group), consistent with poor permeability. (b) BBB-permeable molecules (label 1), Trifluoperidol and Fletazepam, attention weights are concentrated on hydrophobic regions (e.g., trifluoromethyl group), known to enhance membrane penetration. These patterns indicate that Galformer learns chemically relevant substructures associated with BBB permeability.

hydrophobic regions that are relevant to blood–brain barrier (BBB) penetration. In contrast, for the impermeable molecules Lenampicillin and Dihydrostreptomycin, the model assigns close attention to the guanidine group and the carboxylic acid group, which are highly polar and typically associated with poor BBB permeability [83], [84]. Overall, these results suggest that the attention in Galformer can highlight substructures consistent with the properties underlying BBB permeability.

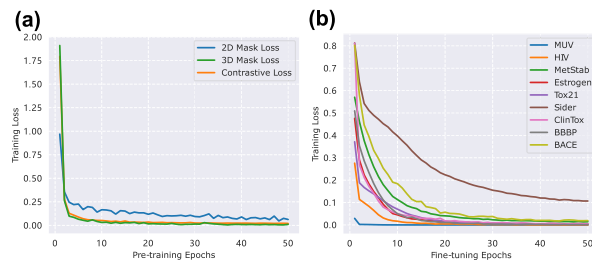


Fig. 11. Convergence analysis of Galformer during pretraining and downstream fine-tuning. (a) Training curves of the 2-D mask loss, 3-D mask loss, and contrastive loss in the pretraining stage. (b) Training curves for the nine downstream tasks during fine-tuning.

F. Convergence Analysis

We conduct a convergence analysis of Galformer during both the pretraining and downstream fine-tuning stages. Fig. 11(a) presents the training curves of the 2-D mask loss, 3-D mask loss, and contrastive loss during pretraining. All three loss components decrease rapidly in the early epochs and then gradually stabilize, indicating fast convergence and stable optimization of the pretraining objectives. Fig. 11(b) illustrates the training loss curves for the nine downstream tasks during fine-tuning. The losses consistently decrease and converge smoothly across all datasets, without significant fluctuations or divergence, demonstrating both reliable convergence speed and strong training stability. Overall, these results indicate that Galformer converges efficiently and stably in both the pretraining and fine-tuning stages.

G. Computational Resource Consumption

To evaluate the computational resources requested by Galformer, we calculate the number of parameters and floating-point operations per second (FLOPs) and compare

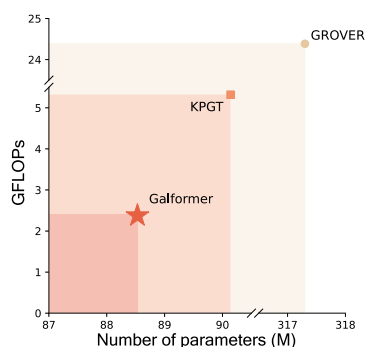


Fig. 12. Comparison of model parameters and FLOPs between Galformer, KPGT, and GROVER. Galformer achieves significantly lower computational cost and model size while maintaining state-of-the-art performance, demonstrating its superior efficiency.

with the other two transformer-based models: KPGT and GROVER. We visualize the results in Fig. 12. The results indicate that Galformer not only achieves SOTA performance on multiple downstream tasks but also has up to ten times less computational complexity and 3.6 times fewer model parameters compared to other graph transformer-based models. It demonstrates the superior efficiency of our model design.

VI. CONCLUSION

In this work, we propose a Galformer, a dual-modality pretraining framework for molecular representation learning. Specifically, we first transform original 2-D and 3-D molecular graphs into line graphs, and then design a dual-modality line graph transformer backbone to effectively encode both topological and geometric information. Furthermore, we introduce two complementary pretraining tasks as self-supervised objectives to capture comprehensive knowledge at both intramodality and intermodality levels. Extensive experiments on downstream tasks demonstrate that Galformer achieves significant improvements over state-of-the-art baselines.

Furthermore, we want to emphasize the potential of Galformer in two future directions: 1) simulating the process of molecule generation and adapting Galformer in structure optimization [85] and 2) extending to more downstream tasks involving large molecules with 3-D information, such as drug-target affinity prediction [86].

REFERENCES

- [1] D. Rogers and M. Hahn, "Extended-connectivity fingerprints," *J. Chem. Inf. Model.*, vol. 50, no. 5, pp. 742–754, May 2010.
- [2] L. Xue and J. Bajorath, "Molecular descriptors in chemoinformatics, computational combinatorial chemistry, and virtual screening," *Combinat. Chem. High Throughput Screening*, vol. 3, no. 5, pp. 363–372, Oct. 2000.
- [3] S. Ji, S. Pan, E. Cambria, P. Marttinen, and P. S. Yu, "A survey on knowledge graphs: Representation, acquisition, and applications," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 2, pp. 494–514, Feb. 2022.
- [4] D. Bacciu and D. Numeroso, "Explaining deep graph networks via input perturbation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 12, pp. 10334–10345, Dec. 2023.
- [5] H. Huang, L. Sun, B. Du, and W. Lv, "Learning joint 2-D and 3-D graph diffusion models for complete molecule generation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 9, pp. 11857–11871, Sep. 2024.
- [6] K. Yang et al., "Analyzing learned molecular representations for property prediction," *J. Chem. Inf. Model.*, vol. 59, no. 8, pp. 3370–3388, Aug. 2019.
- [7] Z. Xiong et al., "Pushing the boundaries of molecular representation for drug discovery with the graph attention mechanism," *J. Medicinal Chem.*, vol. 63, no. 16, pp. 8719–8730, Aug. 2020.

- [8] W. Hu et al., "Deep learning methods for small molecule drug discovery: A survey," *IEEE Trans. Artif. Intell.*, vol. 5, no. 2, pp. 459–479, Feb. 2024.
- [9] Y. Wang, J. Wang, Z. Cao, and A. Barati Farimani, "Molecular contrastive learning of representations via graph neural networks," *Nature Mach. Intell.*, vol. 4, no. 3, pp. 279–287, Mar. 2022.
- [10] Y. Rong et al., "Self-supervised graph transformer on large-scale molecular data," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 12559–12571.
- [11] Z. Zhang, Q. Liu, H. Wang, C. Lu, and C. Lee, "Motif-based graph self-supervised learning for molecular property prediction," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 15870–15882.
- [12] W. Hu et al., "Strategies for pre-training graph neural networks," in *Proc. Int. Conf. Learn. Represent.*, 2020, pp. 20479–20502. [Online]. Available: <https://proceedings.mlr.press/v162/stark22a.html>
- [13] H. Li, D. Zhao, and J. Zeng, "KPGT: Knowledge-guided pre-training of graph transformer for molecular property prediction," in *Proc. 28th ACM SIGKDD Conf. Knowl. Discovery Data Mining*, 2022, pp. 857–867.
- [14] M. Xu, H. Wang, B. Ni, H. Guo, and J. Tang, "Self-supervised graph-level representation learning with local and global structure," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 11548–11558.
- [15] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and P. S. Yu, "A comprehensive survey on graph neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 4–24, Jan. 2021.
- [16] Y. You, T. Chen, Y. Shen, and Z. Wang, "Graph contrastive learning automated," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 12121–12132.
- [17] X. Luo et al., "CLEAR: Cluster-enhanced contrast for self-supervised graph representation learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 1, pp. 899–912, Jan. 2024.
- [18] Y. You, T. Chen, Y. Sui, T. Chen, Z. Wang, and Y. Shen, "Graph contrastive learning with augmentations," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 5812–5823.
- [19] S. Li, X. Wang, A. Zhang, Y. Wu, X. He, and T. Chua, "Let invariant rationale discovery inspire graph contrastive learning," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 13052–13065.
- [20] M. Sun, J. Xing, H. Wang, B. Chen, and J. Zhou, "MoCL: Data-driven molecular fingerprint via knowledge-aware contrastive learning from molecular graph," in *Proc. 27th ACM SIGKDD Conf. Knowl. Discovery Data Mining*, 2021, pp. 3585–3594.
- [21] S. Li, J. Zhou, T. Xu, D. Dou, and H. Xiong, "GeomGCL: Geometric graph contrastive learning for molecular property prediction," in *Proc. AAAI Conf. Artif. Intell.*, Jun. 2022, vol. 36, no. 4, pp. 4541–4549.
- [22] H. Stärk et al., "3D infomax improves GNNs for molecular property prediction," in *Proc. Int. Conf. Mach. Learn.*, 2021.
- [23] J. Zhu et al., "Unified 2D and 3D pre-training of molecular representations," in *Proc. 28th ACM SIGKDD Conf. Knowl. Discovery Data Mining*, 2022, pp. 2626–2636.
- [24] X. Peng, J. Cheng, X. Tang, J. Liu, and J. Wu, "Dual contrastive learning network for graph clustering," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 8, pp. 10846–10856, Aug. 2024.
- [25] S. Liu, H. Wang, W. Liu, J. Lasenby, H. Guo, and J. Tang, "Pre-training molecular graph representation with 3D geometry," in *Proc. Int. Conf. Learn. Represent.*, 2022. [Online]. Available: <https://openreview.net/forum?id=xQe1pOKPam>
- [26] C. Li et al., "Geometry-based molecular generation with deep constrained variational autoencoder," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 4, pp. 4852–4861, Apr. 2024.
- [27] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, "How powerful are graph neural networks?," in *Proc. Int. Conf. Learn. Represent.*, 2019. [Online]. Available: <https://openreview.net/forum?id=ryGs6iA5Km>
- [28] P. Gao et al., "Dynamic fusion with intra- and inter-modality attention flow for visual question answering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6632–6641.
- [29] X. Wei, T. Zhang, Y. Li, Y. Zhang, and F. Wu, "Multi-modality cross attention network for image and sentence matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10938–10947.
- [30] D. Weininger, "SMILES, a chemical language and information system. 1. introduction to methodology and encoding rules," *J. Chem. Inf. Comput. Sci.*, vol. 28, no. 1, pp. 31–36, Feb. 1988.
- [31] H. van de Waterbeemd and E. Gifford, "ADMET in silico modelling: Towards prediction paradise?," *Nature Rev. Drug Discovery*, vol. 2, no. 3, pp. 192–204, Mar. 2003.
- [32] Y. Cao, T. Jiang, and T. Girke, "A maximum common substructure-based algorithm for searching and predicting drug-like compounds," *Bioinformatics*, vol. 24, no. 13, pp. i366–i374, Jul. 2008.

- [33] J. Gilmer, S. S. Schoenholz, P. Riley, O. Vinyals, and G. E. Dahl, "Neural message passing for quantum chemistry," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 1263–1272.
- [34] C. Ying et al., "Do transformers really perform badly for graph representation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 28877–28888.
- [35] E. O. Pyzer-Knapp et al., "Accelerating materials discovery using artificial intelligence, high performance computing and robotics," *npj Comput. Mater.*, vol. 8, no. 1, p. 84, Apr. 2022.
- [36] S. Luo et al., "One transformer can understand both 2D & 3D molecular data," 2022, *arXiv:2210.01765*.
- [37] F. Fuchs, D. Worrall, V. Fischer, and M. Welling, "Se (3)-transformers: 3D roto-translation equivariant attention networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 1970–1981.
- [38] G. Mialon, D. Chen, M. Selsos, and J. Mairal, "Graphit: Encoding graph structure in transformers," 2021, *arXiv:2106.05667*.
- [39] L. Bai et al., "HAQJSK: hierarchical-aligned quantum Jensen–Shannon kernels for graph classification," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 11, pp. 6370–6384, Nov. 2024.
- [40] J. Li, R. Zheng, H. Feng, M. Li, and X. Zhuang, "Permutation equivariant graph framelets for heterophilous graph learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 9, pp. 11634–11648, Sep. 2024.
- [41] K. Choudhary and B. DeCost, "Atomistic line graph neural network for improved materials property predictions," *npj Comput. Mater.*, vol. 7, no. 1, Nov. 2021, Art. no. 185.
- [42] M. Li et al., "Guest editorial: Deep neural networks for graphs: Theory, models, algorithms, and applications," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 4, pp. 4367–4372, Apr. 2024.
- [43] Y. Liu et al., "Graph self-supervised learning: A survey," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 6, pp. 5879–5900, Jun. 2023.
- [44] R. Jiao, J. Han, W. Huang, Y. Rong, and Y. Liu, "Energy-motivated equivariant pretraining for 3D molecular graphs," in *Proc. AAAI Conf. Artif. Intell.*, Jun. 2023, vol. 37, no. 7, pp. 8096–8104.
- [45] J. Feng, Z. Wang, Y. Li, B. Ding, Z. Wei, and H. Xu, "MGMAE: Molecular representation learning by reconstructing heterogeneous graphs with a high mask ratio," in *Proc. 31st ACM Int. Conf. Inf. Knowl. Manage.*, 2022, pp. 509–519.
- [46] X. Yang et al., "A multi-task self-supervised strategy for predicting molecular properties and FGFR1 inhibitors," *Adv. Sci.*, vol. 12, no. 13, Apr. 2025, Art. no. 2412987.
- [47] F.-Y. Sun, J. Hoffmann, V. Verma, and J. Tang, "InfoGraph: Unsupervised and semi-supervised graph-level representation learning via mutual information maximization," in *Proc. Int. Conf. Learn. Represent.*, 2019.
- [48] R. Ma, Y. Zhang, X. Wang, Z. Yu, and L. Duan, "MORN: Molecular property prediction based on textual-topological-spatial multi-view learning," in *Proc. 31st ACM Int. Conf. Inf. Knowl. Manage.*, Oct. 2022, pp. 1461–1470.
- [49] A. Subramonian, "Motif-driven contrastive learning of graph representations," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 18, pp. 15980–15981.
- [50] T. Jiang et al., "Mix-key: Graph mixup with key structures for molecular property prediction," *Briefings Bioinf.*, vol. 25, no. 3, p. 165, Mar. 2024.
- [51] S. Liu, W. Du, Z.-M. Ma, H. Guo, and J. Tang, "A group symmetric stochastic differential equation model for molecule multi-modal pretraining," in *Proc. 40th Int. Conf. Mach. Learn.*, 2023, pp. 21497–21526.
- [52] W. Du, J. Chen, X. Zhang, Z. Ma, and S. Liu, "Molecule joint auto-encoding: Trajectory pretraining with 2D and 3D diffusion," in *Proc. Adv. Neural Inf. Process. Syst.*, 2023, pp. 55077–55096.
- [53] S. Feng et al., "UniCorn: A unified contrastive learning approach for multi-view molecular representation learning," in *Proc. 41st Int. Conf. Mach. Learn.*, 2024, pp. 13256–13277. [Online]. Available: <https://proceedings.mlr.press/v235/feng24f.html>
- [54] X. Wang, H. Zhao, W.-W. Tu, and Q. Yao, "Automated 3D pre-training for molecular property prediction," in *Proc. 29th ACM SIGKDD Conf. Knowl. Discovery Data Mining*, Aug. 2023, pp. 2419–2430.
- [55] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 6000–6010.
- [56] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," 2016, *arXiv:1607.06450*.
- [57] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 770–778.
- [58] X. Fang et al., "Geometry-enhanced molecular representation learning for property prediction," *Nature Mach. Intell.*, vol. 4, no. 2, pp. 1–8, Feb. 2022.
- [59] B. Scholkopf et al., "Comparing support vector machines with Gaussian kernels to radial basis function classifiers," *IEEE Trans. Signal Process.*, vol. 45, no. 11, pp. 2758–2765, Nov. 1997.
- [60] V. P. Dwivedi, C. K. Joshi, A. T. Luu, T. Laurent, Y. Bengio, and X. Bresson, "Benchmarking graph neural networks," 2020, *arXiv:2003.00982*.
- [61] M. Belkin and P. Niyogi, "Laplacian eigenmaps for dimensionality reduction and data representation," *Neural Comput.*, vol. 15, no. 6, pp. 1373–1396, Jun. 2003.
- [62] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," 2018, *arXiv:1810.04805*.
- [63] A. V. D. Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," 2018, *arXiv:1807.03748*.
- [64] S. Axelrod and R. Gómez-Bombarelli, "GEOM, energy-annotated molecular conformations for property prediction and molecular generation," *Scientific Data*, vol. 9, no. 1, p. 185, Apr. 2022.
- [65] Z. Wu et al., "MoleculeNet: A benchmark for molecular machine learning," *Chem. Sci.*, vol. 9, no. 2, pp. 513–530, 2018.
- [66] A. Gaulton et al., "ChEMBL: A large-scale bioactivity database for drug discovery," *Nucleic Acids Res.*, vol. 40, no. D1, pp. D1100–D1107, Jan. 2012.
- [67] S. Podlowska and R. Kafel, "MetStabOn—Online platform for metabolic stability predictions," *Int. J. Mol. Sci.*, vol. 19, no. 4, p. 1040, Mar. 2018.
- [68] F.-J. Gamito et al., "Thousands of chemical starting points for antimalarial lead identification," *Nature*, vol. 465, no. 7296, pp. 305–310, May 2010.
- [69] J. Hachmann et al., "The Harvard clean energy project: Large-scale computational screening and design of organic photovoltaics on the world community grid," *J. Phys. Chem. Lett.*, vol. 2, no. 17, pp. 2241–2251, Sep. 2011.
- [70] J. Zhu et al., "Dual-view molecular pre-training," in *Proc. 29th ACM SIGKDD Conf. Knowl. Discovery Data Mining*, 2023, pp. 3615–3627.
- [71] H. Li, R. Zhang, Y. Min, D. Ma, D. Zhao, and J. Zeng, "A knowledge-guided pre-training framework for improving molecular representation learning," *Nature Commun.*, vol. 14, no. 1, p. 7568, Nov. 2023.
- [72] A. Paszke et al., "Automatic differentiation in PyTorch," in *Proc. Neural Inf. Process. Syst. Autodiff Workshop*, 2017.
- [73] M. Wang et al., "Deep graph library: A graph-centric, highly-performant package for graph neural networks," 2019, *arXiv:1909.01315*.
- [74] G. Landrum. (2006). *Rdkit: Open-Source Cheminformatics*. [Online]. Available: <https://www.rdkit.org/>
- [75] M. Li et al., "DGL-LifeSci: An open-source toolkit for deep learning on graphs in life science," *ACS Omega*, vol. 6, no. 41, pp. 27233–27238, Oct. 2021.
- [76] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, no. 85, pp. 2825–2830, 2011.
- [77] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
- [78] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16000–16009.
- [79] T. Chen, S. Kornblith, M. Norouzi, and G. E. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. 37th Int. Conf. Mach. Learn.*, vol. 119, 2020, pp. 1597–1607.
- [80] L. V. D. Maaten and G. E. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 86, pp. 2579–2605, 2008.
- [81] D. L. Davies and D. W. Bouldin, "A cluster separation measure," *IEEE Trans. Pattern Anal. Mach. Intell.*, vols. PAMI-1, no. 2, pp. 224–227, Apr. 1979.
- [82] W. Zhu, Y. Zhang, D. Zhao, J. Xu, and L. Wang, "HiGNN: A hierarchical informative graph neural network for molecular property prediction equipped with feature-wise attention," *J. Chem. Inf. Model.*, vol. 63, no. 1, pp. 43–55, Jan. 2023.
- [83] K. M. Mahar Doan, S. S. Lakhman, and K. M. K. Boje, "Blood-brain barrier transport studies of organic guanidino cations using an in situ brain perfusion technique," *Brain Res.*, vol. 876, nos. 1–2, pp. 141–147, Sep. 2000.
- [84] S. A. Hitchcock and L. D. Pennington, "Structure-brain exposure relationships," *J. Medicinal Chem.*, vol. 49, no. 26, pp. 7559–7583, 2006.
- [85] M. Sun, J. Xing, H. Meng, H. Wang, B. Chen, and J. Zhou, "Molsearch: Search-based multi-objective molecular generation and property optimization," in *Proc. 28th ACM SIGKDD Conf. Knowl. Discovery Data Mining*, 2022, pp. 4724–4732.
- [86] K. Huang, C. Xiao, L. M. Glass, and J. Sun, "MolTrans: Molecular interaction transformer for drug–target interaction prediction," *Bioinformatics*, vol. 37, no. 6, pp. 830–836, May 2021.



Peizhen Bai received the B.S. degree in information management and information systems from Sun Yat-sen University, Guangzhou, China, in 2016, and the M.S. degree in data analytics and the Ph.D. degree in computer science from the University of Sheffield, Sheffield, U.K., in 2017 and 2025, respectively.

He is currently a Visiting Researcher at the University of Sheffield, and a Senior AI Scientist at AstraZeneca, Cambridge, U.K. His current research interests include AI for life science, protein design, drug discovery, deep learning, graph neural networks, and generative models.



Xianyuan Liu received the B.S. degree in measuring control technology and instruments from Southeast University, Nanjing, China, in 2016, and the Ph.D. degree in signal and information processing from the University of Chinese Academy of Sciences, Beijing, China, in 2023.

He was a Senior AI Research Engineer at the University of Sheffield, Sheffield, U.K., from 2023 to 2026. He is currently an Academic Fellow of multimodal AI with the School of Computer Science and the Centre for Machine Intelligence, University of Sheffield. His current research interests include multimodal learning, deep learning, AI for science, digital materials design, and drug discovery.



Wenrui Fan received the B.S. degree in aerospace engineering from Beijing Institute of Technology, Beijing, China, in 2017, and the M.S. degree in robotics from the University of Sheffield, Sheffield, U.K., in 2022, where he is currently pursuing the Ph.D. degree.

He has been an AI Research Engineer at the University of Sheffield since 2023. His current research interests include multimodal learning, foundation models, AI for science, protein engineering, translational neuroscience, and AI for healthcare.



Tianyi Jiang received the B.E. degree in automation from China Jiliang University, Hangzhou, China, in 2022. She is currently pursuing the Ph.D. degree with the College of Information Engineering, Zhejiang University of Technology, Hangzhou.

From August 2025 to February 2026, she was a Visiting Research Student with the School of Computer Science, University of Sheffield, Sheffield, U.K. Her current research interests include machine learning, bioinformatics, and multimodal AI.



William K. Cheung (Member, IEEE) received the Ph.D. degree in computer science from The Hong Kong University of Science and Technology, Hong Kong, in 1999.

He is currently a Professor with the Department of Computer Science, Hong Kong Baptist University, Hong Kong. His current research interests include artificial intelligence, data mining, collaborative information filtering, social network analysis, and healthcare informatics.

Dr. Cheung has served as the Co-Chair and a Program Committee Member for a number of international conferences and workshops, as well as a Guest Editor of journals in areas including artificial intelligence, web intelligence, data mining, web services, e-commerce technologies, and health informatics. From 2002 to 2018, he was on the Editorial Board of IEEE INTELLIGENT INFORMATICS BULLETIN. He is currently a Track Editor of *Web Intelligence Journal* and an Associate Editor of *Journal of Health Information Research and Network Modeling and Analysis for Health Informatics and Bioinformatics*.



Haiping Lu (Senior Member, IEEE) received the B.Eng. and M.Eng. degrees in electrical and electronics engineering from Nanyang Technological University, Singapore, in 2001 and 2004, respectively, and the Ph.D. degree in electrical and computer engineering from the University of Toronto, Toronto, ON, Canada, in 2008.

He is currently a Professor of machine learning at the School of Computer Science and the Head of AI research engineering at the Centre for Machine Intelligence, University of Sheffield, Sheffield, U.K.

He is also the Director of UK Open Multimodal AI Network (UKOMAIN). He leads the open-source project PyKale for knowledge-aware machine learning, the only multimodal training library in the PyTorch Landscape. His research focuses on deployment-centric multimodal AI, integrating diverse types of data to tackle challenges in healthcare and scientific discovery, with methodological interests spanning foundation models, generative AI, domain adaptation, and transfer learning. His recent work spans brain and cardiac imaging, cancer diagnosis, protein design, and drug and materials discovery.

Dr. Lu received awards from the Alan Turing Institute, Amazon, the Wellcome Trust, and U.K.'s National Institute for Health and Care Research. He served as an Associate Editor for IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS and IEEE TRANSACTIONS ON COGNITIVE AND DEVELOPMENTAL SYSTEMS.